ANSI SQL Using MySQL Exercises[10-15]

11. Daily New User Count

    Find the number of users who registered each day in the last 7 days

```sql
SELECT registration_date,
       COUNT(*) AS total_users
FROM Users
WHERE registration_date >= CURDATE() - INTERVAL 7 DAY
GROUP BY registration_date
ORDER BY registration_date;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

registration_date	total_users

12. Event with Maximum Sessions

    List the event(s) with the highest number of sessions.

```sql
SELECT e.title,
       COUNT(s.session_id) AS total_sessions
FROM Events e
JOIN Sessions s
    ON e.event_id = s.event_id
GROUP BY e.event_id, e.title
HAVING COUNT(s.session_id) = (
    SELECT MAX(session_count)
    FROM (
        SELECT COUNT(*) AS session_count
        FROM Sessions
        GROUP BY event_id
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

title	total_sessions
Tech Innovators Meetup	2

13. Average Rating per City

Calculate the average feedback rating of events conducted in each city

```sql
 -- 13. Average Rating per City
SELECT e.city,AVG(f.rating) as 'average_rating_per_city'
from events e join feedback f on f.event_id = e.event_id
GROUP BY e.city;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

city	average_rating_per_city
New York	4.0769
Chicago	4.5000

14. Most Registered Events

List top 3 events based on the total number of user registrations

```sql
SELECT e.title,
       COUNT(r.registration_id) AS registrations
FROM Events e
JOIN Registrations r
    ON e.event_id = r.event_id
GROUP BY e.event_id, e.title
ORDER BY registrations DESC
LIMIT 3;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

title	registrations
Tech Innovators Meetup	2
AI & ML Conference	2
Frontend Development Bootcamp	1

15. Event Session Time Conflict

Identify overlapping sessions within the same event (i.e., session start and end times that conflict)

```sql
-- 15. Event Session Time Conflict
SELECT s1.event_id,
       s1.session_id,
       s2.session_id
FROM Sessions s1
JOIN Sessions s2
    ON s1.event_id = s2.event_id
   AND s1.session_id < s2.session_id
   AND s1.start_time < s2.end_time
   AND s1.end_time > s2.start_time;
```